

# Chapter 1

## Section 1 - Intro to Logic and Proofs

### 1.1 Section 1.1 - Intro to Logic

#### 1.1.1 Propositions

##### Definition 1 (Proposition)

A **proposition** is a statement that is either

##### Example 1 (Examples with Propositions)

Determine if the following are propositions

1. It's all coming back to me now

*Solution:*

yes

2. Hello.

*Solution:*

No

3.  $x + 3 = 10$

***Solution:***

No, since we don't know  $x$

4. In a right triangle,  $a^2 + b^2 = c^2$

***Solution:***

No, since  $a, b, c$  are not assigned anything

5. In a right triangle having legs of length  $a$  and  $b$ , and hypotenuse of  $c$ , then  $a^2 + b^2 = c^2$

***Solution:***

Yes, since  $a, b, c$  are properly specified

6. There are 75 people in this room or the sky is purple

***Solution:***

Yes

**Note**

7 is a **compound proposition**

**Definition 2 (Compound Proposition)**

A **compound proposition** is a proposition that contains two or more **atomic** propositions. In the previous example, the two are:

1. There are 75 people in this room
2. The sky is purple

**Definition 3 (Atomic Proposition)**

An **atomic proposition** is a proposition that cannot be broken down further into separate propositions

**Note**

We denote separate propositions as:  $p, q, r$

### 1.1.2 Logical Operations (connectives)

| Name               | Notation     | Phrasing               | Truth Value                                     |
|--------------------|--------------|------------------------|-------------------------------------------------|
| Disjunction (or)   | $p \vee q$   | "p or q"               | True when $p, q$ or both are <b>TRUE</b>        |
| Conjunction (and)  | $p \wedge q$ | "p and q"              | True when both are <b>TRUE</b>                  |
| Negation (not)     | $\neg p$     | "not p"                | The opposite truth value of $p$                 |
| Exclusive Or (XOR) | $p \oplus q$ | "p or q, but not both" | <b>TRUE</b> when <b>only one</b> is <b>TRUE</b> |

Here are the truth tables of each operation

| $p$      | $q$      | $p \vee q$ | $p \wedge q$ | $\neg p$ | $p \oplus q$ |
|----------|----------|------------|--------------|----------|--------------|
| <b>T</b> | <b>T</b> | <b>T</b>   | <b>T</b>     | <b>F</b> | <b>F</b>     |
| <b>T</b> | <b>F</b> | <b>T</b>   | <b>F</b>     | <b>F</b> | <b>T</b>     |
| <b>F</b> | <b>T</b> | <b>T</b>   | <b>F</b>     | <b>T</b> | <b>T</b>     |
| <b>F</b> | <b>F</b> | <b>F</b>   | <b>F</b>     | <b>T</b> | <b>F</b>     |

#### Example 2 (Logical Operators Examples)

Let  $p$  be the proposition "a is odd" and  $q$  be "a is even" where  $a = 3$

$p$ : 3 is odd = **TRUE**

$q$ : 9 is even = **FALSE**

Write the following Propositions:

1.  $\neg p$  (3 is even)

**Solution:**

**FALSE**

2.  $\neg q$  (9 is odd)

**Solution:**

**TRUE**

3.  $p \vee q$  (3 is odd or 9 is even)

**Solution:**

**TRUE** since 3 is odd

4.  $p \wedge q$  (3 is odd and 9 is even)

**Solution:**

**FALSE** since 9 is not even

5.  $p \wedge \neg q$  (3 is odd and 9 is odd)

**Solution:**

**TRUE** since both 3 is odd and 9 is odd

### 1.1.3 Conditionals

| Name          | Notation              | Phrasing                       | Truth Value                               |
|---------------|-----------------------|--------------------------------|-------------------------------------------|
| Conditional   | $p \Rightarrow q$     | "p implies q" or "if p then q" | Always True except when $T \Rightarrow F$ |
| Biconditional | $p \Leftrightarrow q$ | "p if and only if q"           | True when $p$ and $q$ are the same value  |

Here are the truth tables

| $p$ | $q$ | $p \Rightarrow q$ | $p \Leftrightarrow q$ |
|-----|-----|-------------------|-----------------------|
| T   | T   | T                 | T                     |
| T   | F   | F                 | F                     |
| F   | T   | T                 | F                     |
| F   | F   | T                 | T                     |

### 1.1.4 Logical Transforms

| Name                                | Original          | Output                      |
|-------------------------------------|-------------------|-----------------------------|
| Converse (swap)                     | $p \Rightarrow q$ | $q \Rightarrow p$           |
| Inverse (negate)                    | $p \Rightarrow q$ | $\neg p \Rightarrow \neg q$ |
| Contrapositive (converse + inverse) | $p \Rightarrow q$ | $\neg q \Rightarrow \neg p$ |

Here are the truth tables

| $p$   | $q$   | $p \Rightarrow q$ | $\neg q$ | $\neg p$ | $\neg q \Rightarrow \neg p$ |
|-------|-------|-------------------|----------|----------|-----------------------------|
| TRUE  | TRUE  | TRUE              | FALSE    | FALSE    | TRUE                        |
| TRUE  | FALSE | FALSE             | TRUE     | FALSE    | FALSE                       |
| FALSE | TRUE  | TRUE              | FALSE    | TRUE     | TRUE                        |
| FALSE | FALSE | TRUE              | TRUE     | TRUE     | TRUE                        |

#### Proposition 1 (Contrapositive Correlation)

From the truth table, we see that the converse is logically equivalent to the original expression, so

$$p \Rightarrow q \equiv \neg q \Rightarrow \neg p$$

**Definition 4 (Logical Equivalence)**

Two propositions are **logically equivalent** if they have the same truth tables

**Example 3 (Logical Transforms Examples)**

Let  $p$  be the proposition "it's cold outside" and  $q$  be "it's snowing outside"

1.  $p \implies q$  (if it's cold outside, then it's snowing outside)

**Solution:**

FALSE

2.  $q \implies p$  (converse) (if it's snowing outside, then it's cold outside)

**Solution:**

TRUE

3.  $\neg p \implies \neg q$  (inverse) (if it's not cold outside, then it's not snowing)

**Solution:**

TRUE

4.  $\neg q \implies \neg p$  (contrapositive) (if it's not snowing outside, then it's not cold outside)

**Solution:**

FALSE

**Example 4 (More Examples with Logical Transforms)**

Let  $p$  be the proposition "The triangle is equilateral" and  $q$  be "The triangle is equiangular"

1.  $p \implies q$  (if the triangle is equilateral, then triangle is equiangular)

**Solution:**

TRUE

2.  $q \implies p$  (converse) (if the triangle is equiangular, then the triangle is

equilateral)

**Solution:**

TRUE

3.  $\neg p \implies \neg q$  (inverse) (if the triangle is not equiangular, then the triangle is not equilateral)

**Solution:**

TRUE

4.  $\neg q \implies \neg p$  (contrapositive) (if the triangle is not equiangular then the triangle is not equilateral)

**Solution:**

TRUE

### Example 5 (Truth Table Examples)

1.  $\neg p \vee q$

| $p$ | $q$ | $\neg p$ | $\neg p \vee q$ |
|-----|-----|----------|-----------------|
| T   | T   | F        | T               |
| T   | F   | F        | F               |
| F   | T   | T        | T               |
| F   | F   | T        | T               |

2.  $(p \implies r) \wedge (q \implies r)$

im too lazy to fill this out

## 1.3 Section 1.3 - Logical Equivalence

### Recall 1 (Logical Equivalence)

Recall that two propositions are **logically equivalent** if they have the same truth table

### 1.3.1 Testing for Equivalence using Truth Tables

#### Example 6

Determine if the following are logically equivalent

1.  $p \implies q$  and  $\neg p \vee q$

*Solution:*

| $p$ | $q$ | $p \implies q$ | $\neg p$ | $\neg p \vee q$ |
|-----|-----|----------------|----------|-----------------|
| T   | T   | T              | F        | T               |
| T   | F   | F              | F        | F               |
| F   | T   | T              | T        | T               |
| F   | F   | T              | T        | T               |

$$\therefore p \implies q \equiv \neg p \vee q$$

2.  $\neg(p \wedge q)$  and  $\neg p \vee \neg q$

*Solution:*

| $p$ | $q$ | $p \wedge q$ | $\neg(p \wedge q)$ | $\neg p$ | $\neg q$ | $\neg p \vee \neg q$ |
|-----|-----|--------------|--------------------|----------|----------|----------------------|
| T   | T   | T            | F                  | F        | F        | F                    |
| T   | F   | F            | T                  | F        | T        | T                    |
| F   | T   | F            | T                  | T        | F        | T                    |
| F   | F   | F            | T                  | T        | T        | T                    |

$$\therefore \neg(p \wedge q) \equiv \neg p \vee \neg q$$

3.  $(p \implies q) \vee (q \implies r)$  and  $p \implies (q \vee r)$

*Solution:*

| $p$ | $q$ | $r$ | $p \implies r$ | $q \implies r$ | $(p \implies r) \vee (q \implies r)$ | $q \vee r$ | $p \implies (q \vee r)$ |
|-----|-----|-----|----------------|----------------|--------------------------------------|------------|-------------------------|
| T   | T   | T   | T              | T              | T                                    | T          | T                       |
| T   | T   | F   | F              | F              | F                                    | T          | T                       |
| T   | F   | T   | T              | T              | T                                    | T          | T                       |
| T   | F   | F   | F              | T              | T                                    | F          | F                       |
| F   | T   | T   | T              | T              | T                                    | T          | T                       |
| F   | T   | F   | T              | F              | T                                    | T          | T                       |
| F   | F   | T   | T              | T              | T                                    | T          | T                       |
| F   | F   | F   | T              | T              | T                                    | F          | T                       |

$$\therefore (p \implies q) \vee (q \implies r) \not\equiv p \implies (q \vee r)$$

So far we know the following:

- $p \implies q \equiv \neg p \vee q$
- DeMorgan's Law
  - $\neg(p \vee q) \equiv \neg p \wedge \neg q$
  - $\neg(p \wedge q) \equiv \neg p \vee \neg q$
- Commutative (swap places)
  - $p \vee q \equiv q \vee p$
  - $p \wedge q \equiv q \wedge p$
- Associative (swap grouping)
  - $(p \vee q) \vee r \equiv p \vee (q \vee r)$
  - $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
- Distributive (distributing operations)
  - $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
  - $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
- Identity
  - $p \wedge T \equiv p$
  - $p \vee F \equiv p$
- Domination Laws
  - $p \vee T \equiv T$
  - $p \wedge F \equiv F$
- Impotent Laws
  - $p \vee p \equiv p$

$$- p \wedge p \equiv p$$

**Definition 5 (Tautology and Contradiction)**

A proposition that is always **TRUE** is a **tautology**

A proposition that is always **FALSE** is a **contradiction**

**Note**

When proving tautologies or contradictions, you could either:

1. Use Truth Tables
2. Use the Laws from earlier to reduce a compound proposition into **TRUE** or **FALSE**

But it's advised to use Truth Tables, since it's easy to get lost in your work when using the Laws

Also, ALWAYS start by **converting conditionals**

**Example 7 (Examples of Tautologies and Contradictions)**

Determine if the following are tautologies, contradictions, or neither

1.  $(p \wedge q) \implies q$

Proving with Truth Tables

| $p$ | $q$ | $p \wedge q$ | $q$ | $(p \wedge q) \implies q$ |
|-----|-----|--------------|-----|---------------------------|
| T   | T   | T            | T   | T                         |
| T   | F   | F            | F   | T                         |
| F   | T   | F            | T   | T                         |
| F   | F   | F            | F   | T                         |

$\therefore (p \wedge q) \implies q \equiv \text{tautology}$

Proving with Logical Equations

$$\begin{aligned}
 (p \wedge q) \implies q &\equiv \neg(p \wedge q) \vee q && \text{(Convert Conditional)} \\
 &\equiv (\neg p \vee \neg q) \vee q && \text{(DeMorgan's Law)} \\
 &\equiv \neg p \vee (\neg q \vee q) && \text{(Associative)} \\
 &\equiv \neg p \vee T \\
 &\equiv T && \text{(Domination)}
 \end{aligned}$$

$\therefore (p \wedge q) \implies q \equiv \text{tautology}$

2. There's another example but I'm too lazy

**Note**

When proving the **logical equivalence** of two propositions, start with the **MORE COMPLICATED** proposition

**Example 8 (Proving Logical Equivalence of Two Propositions)**

Determine if  $(p \vee q) \implies r \equiv (p \implies r) \wedge (q \implies r)$

$(p \implies r) \wedge (q \implies r)$  seems more complicated, so we'll start with that

$$\begin{aligned}(p \implies r) \wedge (q \implies r) &\equiv (\neg p \vee r) \wedge (\neg q \vee r) && \text{(Convert Conditionals)} \\ &\equiv (\neg p \wedge \neg q) \vee r && \text{(Un-Distribute)} \\ &\equiv \neg(p \vee q) \vee r && \text{(DeMorgan's Law)} \\ &\equiv (p \vee q) \implies r && \text{(Convert back to Conditional)}\end{aligned}$$

$$\therefore (p \vee q) \implies r \equiv (p \implies r) \wedge (q \implies r)$$

## 1.4 Section 1.4 - Propositional Functions and Quantifiers

First, let's start with an example

**Example 9**

Is  $x + 3 = 10$  a proposition?

**Solution:**

No, since  $x$  is not known

However, if  $x$  is known, then  $x + 3 = 10$  is a proposition

The truth value depends on the value of  $x$

**Definition 6 (Propositional Function)**

We call  $x + 3 = 10$  a **propositional function** since the truth value of the proposition depends on the value of  $x$

**Example 10 (Examples of Propositional Functions)**

For the given propositional functions, determine the truth value for the given value of variable(s)

1.  $P(x) : x + 3 = 10$

(a)  $P(6)$

**Solution:**

**FALSE** since  $6 + 3 \neq 10$

(b)  $P(7)$

**Solution:**

**TRUE** since  $7 + 3 = 10$

2.  $P(a, b, c) : a^2 + b^2 = c^2$

(a)  $P(11, 60, 61)$

**Solution:**

**TRUE**

$$11^2 + 60^2 = 121 + 3600 = 3721$$

$$61^2 = (60 + 1)^2 = 3600 + 120 + 1 = 3721$$

(b)  $P(147, 201, 513)$

**Solution:**

**FALSE**

$$147^2 + 201^2 = \text{odd} + \text{odd} = \text{even}$$

$$513^2 = \text{odd} \neq \text{even}$$

**Example 11 (Examples of Notation)** 1. Is "there exists a real number  $x$  such that  $x + 3 = 10$ ?" a proposition?

**Solution:**

Yes, and this proposition is **TRUE** since  $x = 7$  works

We can also rewrite this as:

$$\exists x \in \mathbb{R} \mid x + 3 = 10$$

2. Is "for all real numbers  $x$ ,  $x + 3 = 10$ " a proposition?

**Solution:**

Yes, but the proposition is **FALSE** since  $x = 6$  doesn't work

We can also rewrite this as

$$\forall x \in \mathbb{R} \mid x + 3 = 10$$

**Definition 7 (Quantifiers)**

The **existential quantifier**  $\exists$  means "there exists"

The **universal quantifier**  $\forall$  means "for all" or "for every"

**Example 12 (Examples of Quantifiers)**

Write each proposition using quantifiers, and determine their truth values

1. There exists an integer  $x$  such that  $x^2 + 2 > 3x$

**Solution:**

$$\exists x \in \mathbb{Z} \mid x^2 + 2 > 3x$$

**TRUE** since  $x = 3$  gives  $9 + 2 > 9$

2. For all real numbers  $x$ ,  $x^2 + 2 > 3x$

**Solution:**

$$\forall x \in \mathbb{R} \mid x^2 + 2 > 3x$$

**FALSE** since  $x = 1$  gives  $1 + 2 > 3$

**Note**

We only need one counter example to disprove a proposition that has  $\forall$

3. There exists an integer  $x$  such that  $x^3 = 31687$

**Solution:**

$$\exists x \in \mathbb{Z} \mid x^3 = 31687$$

**FALSE** since  $\sqrt[3]{31687} \notin \mathbb{Z}$

**Note**

We needed to verify that every integer does not satisfy the proposition in order to prove it

The negations of each quantifier are as follows:

$$\neg \forall \equiv \exists$$

$$\neg \exists \equiv \forall$$

**Example 13 (Negation of Quantifiers)**

Give the negation of the following propositions:

1.  $\forall x \in \mathbb{R} \mid x^2 + 1 > 0$  **TRUE**

**Solution:**

$$\exists x \in \mathbb{R} \mid x^2 + 1 \leq 0$$
 **FALSE**

2.  $\exists a \in \mathbb{Z} \mid a^3 + 2 = 3a$  **TRUE** ( $a = 1$ )

**Solution:**

$$\forall a \in \mathbb{Z} \mid a^3 + 2 \neq 3a$$
 **FALSE**

3. For every real number  $x$ ,  $x^2 + 30 < 11x$  **FALSE** ( $x = 1$ )

**Solution:**

$$\exists x \in \mathbb{R} \mid x^2 + 30 \geq 11x$$
 **TRUE**

4. For every prime integer  $p > 3$ ,  $p^2 - 1$  is divisible by 24 **TRUE**

**Solution:**

There exists a prime integer  $p > 3$ ,  $p^2 - 1$  is not divisible by 24 **FALSE**

Work

$$p^2 - 1$$

We want  $p^2 - 1$  divisible by both 8 and 3

$$(p^2 - 1) = (p - 1)(p + 1)$$

$$(p - 1)(p + 1) = (\text{even})(\text{even})$$

- $p$  is prime
- $p > 3$
- $p$  is odd

**Example 14**

| Statement         | When <b>TRUE</b>                                              | When <b>FALSE</b>                                              |
|-------------------|---------------------------------------------------------------|----------------------------------------------------------------|
| $\forall x, P(x)$ | $P(x)$ is <b>TRUE</b> for all $x$                             | There exists at least one $x$ such that $P(x)$ is <b>FALSE</b> |
| $\exists x, P(x)$ | There exists at least one $x$ such that $P(x)$ is <b>TRUE</b> | For all $x$ such that $P(x)$ is <b>FALSE</b>                   |

**Example 15 (Examples with Negation)**

| Statement              | Equivalent             | When <b>TRUE</b>                                           | When <b>FALSE</b>                                             |
|------------------------|------------------------|------------------------------------------------------------|---------------------------------------------------------------|
| $\neg \forall x, P(x)$ | $\exists x, \neg P(x)$ | There is at least one $x$ such that $P(x)$ is <b>FALSE</b> | For all $x$ , $P(x)$ is <b>TRUE</b>                           |
| $\neg \exists x, P(x)$ | $\forall x, \neg P(x)$ | $P(x)$ is <b>FALSE</b> for all $x$                         | There exists at least one $x$ such that $P(x)$ is <b>TRUE</b> |

**Example 16 (Examples of Rewriting with Quantifiers)**

Rewrite the following with Quantifiers

1. For all positive integers, either  $6n + 1$  or  $6n - 1$  is prime

**Solution:**

$\forall n \in \mathbb{Z}^+, 6n - 1$  is prime  $\vee 6n + 1$  is prime

**FALSE** since if  $n = 20$  then  $6n - 1 = 119$  and  $6n + 1 = 121$  are **composite** (not prime)

2. There exists an integer  $n$  such that  $n^2 + n + 41$  is composite

**Solution:**

$\exists n \in \mathbb{Z}, n^2 + n + 41$  is composite

TRUE since if  $n = 41$  then

$$\begin{aligned} n^2 + n + 41 &= 41^2 + 41 + 41 \\ &= 41(41 + 1 + 1) \end{aligned}$$

which is composite

## 1.5 Section 1.5 - Nested Quantifiers

Let  $\mathbb{P}$  be the set of prime integers

$$\mathbb{P} = \{2, 3, 5, 7, 11, \dots\}$$

With this in mind, we can write the following in two ways:

Consider the statement "for all positive integers  $n$ ,  $6n - 1$  or  $6n + 1$  is prime"

1.  $\forall n \in \mathbb{Z}^+, 6n - 1 \in \mathbb{P} \vee 6n + 1 \in \mathbb{P}$
2.  $\forall n \in \mathbb{Z}^+ \exists p \in \mathbb{P}, 6n - 1 = p \vee 6n + 1 = p$

**Example 17 (Example of Nested Quantifiers)** 1. For all real  $x$  there exists a real number  $y$  such that  $xy = 1$

$$\forall x \in \mathbb{R} \exists y \in \mathbb{R}, xy = 1$$

FALSE, since if  $x = 0$ , then  $xy = 0 \neq 1$

2. For all positive real  $x$ , there exists a real number  $y$  such that  $xy = 1$

$$\forall x \in \mathbb{R}^+ \exists y \in \mathbb{R}, xy = 1$$

TRUE since if  $x \in \mathbb{R}^+$ , then  $x > 0$ , and if  $y = \frac{1}{x}$ , then  $xy = x \frac{1}{x} = 1$

**Note**

Given  $x$ , we can find  $y$

$y$  depends on  $x$

But what happens if we switch  $\forall$  and  $\exists$ ?

3.  $\exists y \in \mathbb{R} \forall x \in \mathbb{R}^+, xy = 1$

**FALSE** since if  $x = 1$ , then  $y = 1$ , but if  $x = 2$ , then  $y = \frac{1}{2}$ , and  $1 \neq \frac{1}{2}$

**Note**

There is a single  $y$  that works **FOR ALL**  $x$

$y$  does not depend on  $x$

4. For all positive integers  $a$  and  $b$ , we have  $ab > 0$

$$\forall a \in \mathbb{Z}^+ \forall b \in \mathbb{Z}^+, ab > 0$$

**Note**

This is the same as  $\forall a, b \in \mathbb{Z}^+$

**TRUE** since  $a, b \in \mathbb{Z}^+$ , then  $a > 0$  and  $b > 0$ , hence  $ab > 0$

5. There exists integers  $a$ , and  $b$  such that  $a^2 + b^2 = 464$

$$\exists a \in \mathbb{Z} \exists b \in \mathbb{Z}, a^2 + b^2 = 464$$

$$\exists b \in \mathbb{Z} \exists a \in \mathbb{Z}, a^2 + b^2 = 464$$

**TRUE**  $a = 8, b = 20$ ,

$$\begin{aligned} a^2 + b^2 &= 8^2 + 20^2 \\ &= 64 + 400 \\ &= 464 \end{aligned}$$

**Note**

The order of the **SAME** quantifier doesn't matter, only for **DIFFERENT** quantifiers does it matter

**Example 18**

Rewrite the following using Quantifiers

1. "There exists positive integers  $a$  and  $b$  such that  $a^2 + b^2 = 15625$

(a) Write as Quantifier

$$\exists a \in \mathbb{Z}^+ \exists b \in \mathbb{Z}^+, a^2 + b^2 = 15625$$

(b) Write the negation

$$\forall a \in \mathbb{Z}^+ \forall b \in \mathbb{Z}^+, a^2 + b^2 \neq 15625$$

*Work*

$$a^2 + b^2 = 5^6$$

$$a^2 + b^2 = 5^4 \cdot 5^2$$

$$3^2 + 4^2 = 5^2$$

$$5^4(3^2 + 4^2) = 5^4 \cdot 5^2$$

$$5^4 \cdot 3^2 + 5^4 \cdot 4^2 = 5^6$$

$$(5^2 \cdot 3)^2 + (5^2 \cdot 4)^2 = 5^6$$

$$(75)^2 + (100)^2 = 5^6$$

$$\therefore a = 75, b = 100$$

TRUE

There's more examples but I'm just too lazy to write them if you get the chance, they're on page 3 of your notes

## 1.7 Section 1.7 Intro to Proofs

What do we prove? Usually that some pattern or formula is true.

### Example 19

$$\begin{aligned}
6 + 8 &= 14 \\
12 + 16 &= 28 \\
2 + 2 &= 4
\end{aligned}$$

We believe that the sum of 2 even integers is even.

Try to prove that: even + even = even

*Work* An even number can be defined as  $2n$ , where  $n$  is an integer

$$2n + 2m = 2(n + m)$$

*Proof.* let  $a, b$  be even integers.

Then there exists integers  $n, m$  such that  $a = 2n, b = 2m$

$$a + b = 2n + 2m = 2(m + n)$$

is even, as desired

□

But writing it this way is too long. Let's define some notation:

### 1.7.1 Set Notation

| Name                       | Symbol         | Set Notation                                         |
|----------------------------|----------------|------------------------------------------------------|
| Set of all natural numbers | $\mathbb{N}$   | $\{0, 1, 2, 3, \dots\}$                              |
| Set of all integers        | $\mathbb{Z}$   | $\{0, -1, 1, -2, 2, \dots\}$                         |
| Set of positive integers   | $\mathbb{Z}^+$ | $\{1, 2, 3, \dots\}$                                 |
| Set of negative integers   | $\mathbb{Z}^-$ | $\{-1, -2, -3, \dots\}$                              |
| Set of rational numbers    | $\mathbb{Q}$   | $\{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\}$ |
| Set of real numbers        | $\mathbb{R}$   | Limit of all convergent sequences of real numbers    |
| Set of all complex numbers | $\mathbb{C}$   | $\{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}$      |

$\exists$  means there exists

Let's rewrite the previous example using this new notation

$a, b \in \mathbb{Z}$ , both even

Then,  $\exists m, n \in \mathbb{Z}$  such that

$$a = 2m, b = 2n$$

Their sum  $a + b = 2m + 2n = 2(m + n)$

is even, as desired

### 1.7.2 Methods of Proving $p \implies q$

1. **Direct Proof:**

Start with  $p$  and *math* then conclude  $q$

2. **Indirect Proof:**

Prove the contrapositive proof  $\neg q \implies \neg p$

3. **Finite Cases:**

Use the fact that  $p \equiv p_1 \vee p_2 \vee \dots \vee p_n$

Then prove  $(p_1 \implies q) \wedge (p_2 \implies q) \wedge \dots \wedge (p_n \implies q)$